

PROFESSIONAL PROJECT PROPOSAL

ClearSight

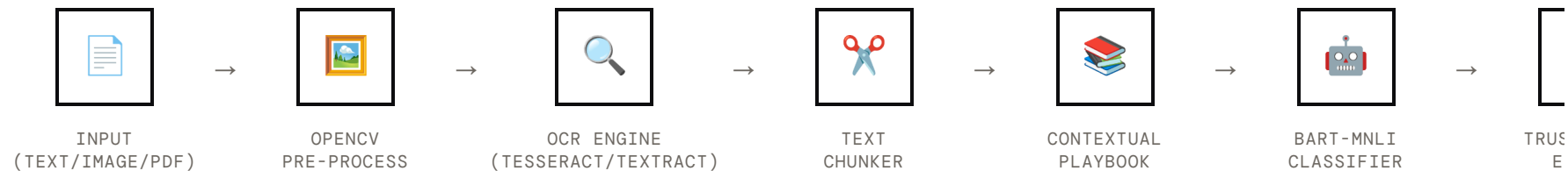
A full-stack Legal Intelligence platform eliminating information asymmetry — turning opaque contracts into clear, proactive, actionable intelligence for individuals and small businesses worldwide.

VERSION	PHASES	TIMELINE	STACK	TARGET
v2.0	3 Phases	10 Weeks	React · Node · BART · OCR	Individuals & SMBs
EDGE				
Proactive · Physical · Accessible				

00 — SYSTEM ARCHITECTURE

How ClearSight thinks

Before writing a single line of code, memorise this pipeline. Every architectural decision traces back to this flow.



01 — COMPETITIVE LANDSCAPE

Who else is in the room

In 2026, the market splits between Enterprise Giants and Niche Add-ons.

ClearSight sits in the gap neither serves: proactive, accessible, and physical-world capable.

COMPETITOR	WHAT THEY DO	HOW THEY DO IT	CLEARLIGHT OPPORTUNITY
Ironclad / Evisort	Enterprise Contract Lifecycle Management	Deep Salesforce integration; massive LLMs trained on legal data	<i>\$10k+ price point. "Overkill" for individuals. Zero accessibility for the average person.</i>
Spellbook	AI Copilot for lawyers	Microsoft Word Add-in that suggests language during drafting	<i>Locked into Word. Built for lawyers drafting, not consumers understanding. No OCR or mobile.</i>
ToS;DR	Crowdsourced T&S ratings	Human volunteers manually read and rate Big Tech policies	<i>Not real-time. Only covers major companies — useless for your local gym or a new startup's contract.</i>
ChatPDF / Claude	General document Q&A	Upload a PDF and "chat" with it to find information	<i>Reactive, not proactive. The user must know what to ask. Doesn't automatically flag risk.</i>
LegalSifter	AI + human legal review	High-end service for corporate legal departments	<i>Slow turnaround vs. instant AI. B2B only — entirely ignores the consumer market.</i>

The ClearSight Edge

ClearSight is the only tool that is simultaneously

Proactive

(automatically flags risks, no prompting required),

Accessible

(web and mobile-first, zero cost barrier), and

Physical

(OCR for paper contracts, not just digital files). It is the "CamScanner" of the legal world — built for the person, not the corporation.

02 — STRATEGIC VALUE

The impact ClearSight creates



Empowerment



Economic Utility



Data Potential

Closing the information asymmetry gap between a company with 100 lawyers and a person signing alone at a kitchen table. ClearSight is a legal equaliser.

Reducing legal friction for small businesses who cannot afford a \$500/hr retainer for every vendor contract, lease renewal, or freelance agreement.

Aggregating Trust Scores across users creates a "Yelp for Ethics" — a community database showing which companies have the fairest (and most predatory) contracts.



Global Reach

Tesseract supports 100+ languages. The same architecture that analyses an English NDA can analyse a Spanish lease or an Arabic employment contract with a config change.

Every tool, every role, every reason

Each technology was chosen deliberately. Do not swap components without understanding the trade-off — the stack is designed as a system.

FRONTEND FRAMEWORK

React + Vite

Fast, component-driven UI with near-instant hot-reloads. Use functional components and hooks exclusively. Vite's build speed means you iterate in seconds, not minutes.

FRONTEND

STATE MANAGEMENT

Context API → Zustand

Start with React Context API for simplicity. Migrate to Zustand if state gets complex across 3+ views. Manage the multi-step pipeline state machine: Idle → Uploading → Processing → Analyzing → Done.

FRONTEND

STYLING

Tailwind CSS

Utility-first framework. Build the enterprise legal aesthetic with responsive prefixes (sm:, md:, lg:) from day one. Never write a custom CSS file until you genuinely need to override Tailwind.

FRONTEND

BACKEND RUNTIME

Node.js + Express

The central orchestrator. Receives all user data, calls OCR, sends text to HF, formats and returns responses. Acts as a secure API

FILE HANDLING

Multer + pdf-parse

Multer handles multipart binary uploads (images, PDFs). pdf-parse extracts clean text from digital PDFs page by page. Together they

VALIDATION

Zod

Define strict schemas for every API payload. Reject invalid file types, enforce text length limits, and catch malformed requests

proxy — your credentials never reach the client.

BACKEND

IMAGE PRE-PROCESSING

OpenCV.js / Sharp.js

Deskew, greyscale, normalise contrast, and sharpen edges before OCR. This single pre-processing step lifts Tesseract accuracy from ~65% on real photos to 90%+. Non-optional for the CamScanner feature.

OCR PIPELINE

AI CLASSIFIER (MVP)

BART-Large-MNLI

Zero-shot classification via Hugging Face Inference API. Define labels (Privacy Risk, Hidden Fee, Liability, Arbitration Clause) and the model scores every text chunk. No training data required — the core enabler of the MVP.

handle every non-text input type in Phase 2 and 3.

BACKEND

OCR — FREE TIER

Tesseract.js

JavaScript port of Google's Tesseract. Runs in Node.js, free, offline-capable, 100+ language support. Primary OCR engine. Route to Textract only when Tesseract confidence falls below 70%.

OCR PIPELINE

AI CLASSIFIER (SCALE)

Legal-BERT

Pre-trained on legal case law and contracts. Understands "legalese" far better than a general model. Upgrade path from BART-MNLI for Phase 3 when you need clause-level precision on professional contracts.

before they reach your AI pipeline. Prevents silent failures.

BACKEND

OCR — PRODUCTION

AWS Textract

Production-grade OCR with table and form detection. Handles dense multi-column legal documents that defeat Tesseract. 1,000 free pages/month. Fallback only — keeps costs near zero in development.

OCR PIPELINE

CLIENT-SIDE SECURITY

DOMPurify + Immediate Deletion

Sanitize all text inputs client-side with DOMPurify before sending to backend. Implement server-side immediate deletion of uploaded images and PDFs after

AI / ML

AI / ML

processing — never persist user documents.

SECURITY

PDF EXPORT

jsPDF

Generate downloadable analysis reports from the frontend. Render the Trust Score, summary table, and full risk card details into a professional PDF the user can share with a lawyer or keep on file.

FRONTEND

DATABASE (PHASE 3)

Supabase (PostgreSQL)

Hosted Postgres with a REST API and auth built in. Store anonymized Trust Scores per company for the community database. Free tier handles the entire Phase 3 build with room to spare.

BACKEND

HOSTING

Vercel + Render

Vercel for the React frontend (zero-config deployment, global CDN). Render or DigitalOcean for the Express backend. Connect both to GitHub for automatic deploys on every push to main.

DEVOPS

⚠ SCALING TIP — WHEN TO UPGRADE FROM BART TO LEGAL-BERT

BART-MNLI is your MVP workhorse. Switch to **Legal-BERT** in Phase 3 when you start handling professional contracts (NDAs, leases, employment agreements). BART understands language broadly; Legal-BERT understands the specific vocabulary, sentence structures, and nuance of legal documents — the difference between "best efforts" and "reasonable

efforts" in a contract matters enormously, and Legal-BERT knows that. The migration is a single model-endpoint swap in your backend config.

04 — EXECUTION ROADMAP



DIGITAL MVP

Weeks 1 - 3

The Digital MVP

Ship a working web-based T&C analyzer — paste text, receive color-coded risk cards — that real users can access by Week 3.

W1

Infrastructure & Project Skeleton

Scaffold Vite + React (client/) and Express (server/) in a monorepo. Configure CORS, connect with Axios, wire up .env for secrets. Set up GitHub with main/dev/feature-* branches. Add ESLint + Prettier with format-on-save. Push a health-check endpoint that returns {"status": "ok"} and confirm the frontend can call it. Do not write application logic until both services communicate cleanly.

W2

Hugging Face AI Bridge

Create POST /api/analyze on Express. Write a text-chunking utility that splits input into ~300-word segments with 50-word overlaps (prevents cutting mid-clause). Call the

BART-Large-MNLI Inference API with labels: ["Privacy Risk", "Hidden Fee", "Auto-Renewal Trap", "Liability Waiver", "Arbitration Clause", "Fair"]. Build and test the entire chain in Postman before touching the frontend — this is backend-only week.

Pro-tip: Save your Postman collection as a JSON file in the repo under /docs/api-tests.json. Future collaborators will thank you.

W3

Analysis Dashboard UI

Build the complete React dashboard: textarea with character counter, "Analyze" button with multi-state loading spinner, and a results zone rendering RiskCard components. Map HF scores to severity — above 0.85 = Red, 0.60–0.85 = Amber, below 0.60 = Blue. Add a Summary Bar showing total flag counts. Handle API errors gracefully (rate limits, timeouts). Deploy to Vercel + Render. Share with 5 people for live feedback before Phase 2.

02

THE VISION LAYER

Weeks 4 - 6

The Vision Layer

Bring ClearSight into the physical world — photograph a paper contract, get an AI analysis — while hardening the platform with security and privacy controls.

W4

OCR Engine Integration

Add a "Scan Document" tab with react-dropzone (drag-and-drop, accepts JPG/PNG/WEBP, max 10MB). On the backend, integrate Tesseract.js via Multer. Create POST /api/ocr that returns extracted text + confidence scores. On mobile, use input type="file" capture="environment" to open the camera directly. Log all sub-70% confidence requests for quality monitoring.

W5

Image Pre-processing Pipeline

Before OCR, run every image through Sharp.js: auto-rotate (EXIF correction), convert to greyscale, normalise contrast, apply unsharp-mask sharpening, output as lossless PNG. Add a basic crop/rotate UI control in React so users can straighten skewed photos before submission. Build the end-to-end route POST /api/scan-and-analyze: Sharp → Tesseract → chunk → BART-MNLI → results. If Tesseract confidence is below 70%, auto-fallback to AWS Textract silently.

OpenCV.js note: For the browser-side crop/rotate tool, OpenCV.js works but is heavy (8MB+). Favour Sharp.js server-side for all processing — it's faster, lighter, and easier to debug in Node.

W6

Security & Privacy Hardening

Implement client-side input sanitization with DOMPurify on all text fields. Server-side: delete all uploaded images and PDFs from disk immediately after processing completes — never persist user documents. Add rate limiting to all routes (express-rate-limit: 20 req/min per IP). Add the mandatory first-time Legal Disclaimer modal in React (stored in localStorage so it only shows once). Add HTTPS enforcement and security headers via Helmet.js.



THE INTELLIGENCE LAYER

Weeks 7 - 10

The Intelligence Layer

Evolve from a T&C scanner into a full contract intelligence platform — with contextual playbooks, a Trust Score engine, PDF support, and community data.

W7

PDF Parsing & Contract-Type Detection

Add PDF upload support via pdf-parse. Extract text per page; route image-based PDF pages through the OCR pipeline. Add a preliminary BART-MNLI classification step that detects the document type (NDA, Lease Agreement, Employment Contract, Service Agreement, General T&C) before the main analysis. This drives the Contextual Playbook selection in Week 8.

W8

Contextual Playbooks

Store label sets as JSON config files — one per contract type. Lease: ["Security Deposit Terms", "Eviction Clause", "Landlord Entry Rights", "Maintenance Responsibility"]. NDA: ["Broad Disclosure Scope", "Perpetual Confidentiality", "IP Assignment"]. Employment: ["Non-Compete Scope", "At-Will Termination", "IP Ownership"]. The backend loads the correct playbook based on the contract-type detection result. Add a manual override dropdown in the UI for user correction.

This is the feature that separates ClearSight from every competitor. Domain-specific label sets mean your accuracy on professional contracts will dramatically exceed any general-purpose tool.

W9

The Trust Score Engine

Build the Trust Index algorithm: start at 100, subtract weighted penalty points per flag — Red = -15 pts, Amber = -5 pts, floored at 0. Display as an animated circular gauge with color transitions: red (0-40), amber (41-70), green (71-100). Add a company name input to the analysis flow. Persist anonymized scores to Supabase. Build a public "Company Lookup" page — search any company name and see its aggregated Trust Index from all community analyses.

W10

One-Click Export & Final Polish

Implement jsPDF export: a professional report with document metadata, Trust Score gauge, severity summary table, and all risk cards with plain-language explanations. Add "What this means for you" copy for every label in the playbook library. Run a final round of user testing across desktop and mobile. Write a README documenting setup,

architecture, and every API route. This is your portfolio submission — treat the documentation as seriously as the code.

05 — PROFESSIONAL GROWTH

What you will master

Every competency below maps directly to a job description in full-stack engineering, AI product development, or technical entrepreneurship.



Full-Stack Architecture

Design a React SPA, REST API, and multiple third-party services as a coherent, scalable system.



AI Integration

Zero-shot classifiers solving a real domain problem. Moving beyond chatbots to specific, useful intelligence.



OCR & Computer Vision

Pre-processing pipelines, confidence routing, and the Sharp → Tesseract → Textract fallback chain.



Async State Management

Multi-step loading, success, and error states across a pipeline spanning 3 external services.



API Security

Secrets server-side, input sanitization, rate limiting, file deletion policies, and Helmet.js headers.



Database Design

Model a community-driven Trust Score database with aggregation, anonymization, and upsert operations.



Mobile-First UX

Camera capture flows on real mobile browsers — a constraint that teaches responsive design at depth.



Deployment & DevOps

Vercel + Render, GitHub-connected auto-deploys, staging vs. production environment separation.

06 — FUTURE VISION

Where ClearSight goes next

Chrome Extension

Real-time risk highlighting on any website's T&C page before the user clicks "I Accept." DOM injection of color-coded overlays, no copy-paste required.

The Trust Index™ Registry

A community-driven database of company Trust Scores aggregated across thousands of analyses. Yelp for contract ethics — searchable and publicly accessible.

Legal-BERT Upgrade

Swap BART-MNLI for Legal-BERT in production to achieve clause-level precision on professional contracts. The architecture already supports this — it's a single config change.

LLM-Powered Chat

After analysis, let users ask: "Can I negotiate clause 7?" or "What is the worst-case scenario here?" — answered by a prompted LLM with the full document as context.

B2B API Product

Package the analysis pipeline as a paid REST API for law firms, HR platforms, and real estate apps to embed natively. This is the monetization path and the natural exit.

Multilingual Contracts

Tesseract's 100+ language support is already in the stack. One config extension opens ClearSight to Spanish, French, and Arabic markets — all deeply underserved by current legal tech.

The disclaimer you must ship

This is not optional. ClearSight is an AI-powered analysis tool, not a law firm. Every first-time user must acknowledge this before using the platform. Store acknowledgement in localStorage so it only displays once per device.

⚠ MANDATORY FIRST-TIME MODAL — SUGGESTED COPY

ClearSight is an AI-powered analysis tool, not legal advice.

ClearSight identifies potentially significant clauses in legal documents using artificial intelligence. The analysis provided is for informational purposes only and does not constitute legal advice, a legal opinion, or an attorney-client relationship. You should consult a qualified legal professional before making any decisions based on this analysis. By continuing, you acknowledge that you understand these limitations.

08 — PRE-BUILD GEAR CHECKLIST

Before you write a single line of code

Every item below must be ticked before starting Week 1. Setup mistakes cost 3× more time to fix later than to prevent now. Your progress saves automatically in this browser.

0 / 10 complete

☐ Hugging Face Account + API Key

Access facebook/bart-large-mnli via the Inference API. Free tier covers all development. Store as HF_API_KEY in server/.env.

☐ Node.js v20+ installed via nvm

Use nvm for version management. Confirm with node --version. Install nvm first if needed – it prevents version conflicts permanently.

☐ Postman or Hoppscotch installed

Non-negotiable. Test every backend route in Postman before writing any frontend code that calls it. Save your collection in /docs/.

☐ GitHub Repository + Branch Strategy

Monorepo: /client, /server, /shared. Enable branch protection on main. Use conventional commits: feat:, fix:, chore:, docs:.

☐ .env configured & .gitignored

Create server/.env with HF_API_KEY, PORT, NODE_ENV. Add .env and .env.local to .gitignore before your first commit. Check with git status.

☐ VS Code + ESLint + Prettier

Configure format-on-save (Prettier) and lint-on-save (ESLint). Consistent style prevents technical debt as the codebase scales across phases.

☐ AWS Account + Textract IAM User

Create an IAM user with AmazonTextractFullAccess. Store access key + secret in .env. Never use root credentials. 1,000 free pages/month.

☐ Vercel + Render accounts linked to GitHub

Connect both platforms to your repo. Configure staging (dev branch) and production (main branch) environments separately from day one.

☐ Supabase Project created (Phase 3 prep)

Free tier provides a hosted Postgres instance. Create the project now, save the connection URL in .env. You will use it in Week 9.

☐ Legal Disclaimer copy written & approved

Draft the first-time user modal text before launch. Have a non-technical person review it for clarity. This is a user-trust and liability protection feature.

ClearSight — Legal & Contract Intelligence Platform

Blueprint v2.0 · 10-Week Build · 3 Phases

Built for individuals & small businesses worldwide